

Assignment 8, Social Science Inquiry II (SOSC13200-W26-3)

Friday 2/27/26 at 11:59pm

This assignment uses the replication materials for:

Offer-Westort, Molly, Will Gruen, Carter Herron, Kaden Hyatt, Max Buford, Kevin Davis, Diego Fonseca, Mushkie Gurevich, Tiffanie Huang, Rocio Jerez, and others. 2026. "Which frame fits? Policy learning with framing for climate change policy attitudes." *Research & Politics*.

Data + code: <https://github.com/UChicago-pol-methods/climate-framing-replication>

What you're practicing this week: treatment effects, covariate adjustment (pre-test control), randomization inference, and power/MDE thinking.

Setup

1. Clone (or download) the replication repo linked above.
2. In that repo, the cleaned dataset lives at `data/ssi-data-cleaned.csv`.
3. Set `data_path` below to point to that file on your computer.

Packages

```
library(ggplot2)
library(stats)
# If you want: install.packages("dplyr"); library(dplyr)
# If you want: install.packages("estimatr"); library(estimatr)
```

Load data

```
data_path <- "https://raw.githubusercontent.com/UChicago-pol-methods/climate-framing-replication/refs/h
dat <- read.csv(data_path, as.is = TRUE)
```

Quick sanity checks (don't delete; you'll use this to orient yourself)

```
dim(dat)
```

```
## [1] 2344 37
```

```
names(dat)
```

```
## [1] "ideology" "scientific_confidence" "reward_consequence"
## [4] "religiosity" "economic_reasoning" "regcarb_after"
## [7] "treatment_value" "age" "party_id"
## [10] "party" "prosociality" "attention_check_1pass"
## [13] "attention_check_2pass" "sex_id" "college"
## [16] "relationship" "income_level" "race_id"
## [19] "race_white" "employment_status" "rel_freq"
## [22] "gastax" "carbtax" "treaty"
## [25] "regcarb" "gastax_after" "carbtax_after"
## [28] "treaty_after" "post_test" "pre_test"
## [31] "unique_id" "treatment_0" "treatment_1"
## [34] "treatment_2" "treatment_3" "treatment_4"
## [37] "treatment_5"
```

```
head(dat)
```

```
## ideology scientific_confidence reward_consequence religiosity
## 1 6 3 1 1
## 2 3 4 1 2
## 3 4 4 1 2
## 4 6 1 1 3
## 5 3 4 1 1
## 6 4 4 2 4
## economic_reasoning regcarb_after treatment_value age party_id party
## 1 2 0 2 43 3 1
## 2 4 2 2 59 0 0
## 3 4 1 5 31 3 1
## 4 5 0 3 52 3 1
## 5 4 3 0 29 0 0
## 6 4 0 5 48 2 1
## prosociality attention_check_1pass attention_check_2pass sex_id college
## 1 1 1 1 1 0
## 2 1 1 1 0 1
## 3 1 1 1 0 0
## 4 1 1 1 0 1
## 5 2 1 1 1 1
## 6 1 1 0 1 1
## relationship income_level race_id race_white employment_status rel_freq
## 1 1 15 7 1 1 1
## 2 0 3 3 0 0 1
## 3 0 2 3 0 0 1
## 4 1 6 7 1 1 3
## 5 0 7 4 0 1 1
## 6 0 2 7 1 0 5
## gastax carbtax treaty regcarb gastax_after carbtax_after treaty_after
## 1 0 0 0 0 0 0 0
## 2 1 2 2 2 1 2 2
## 3 0 1 2 2 0 0 2
## 4 0 0 0 0 0 0 0
## 5 3 3 3 3 3 3 3
## 6 0 0 0 0 0 0 0
## post_test pre_test unique_id treatment_0 treatment_1 treatment_2 treatment_3
## 1 0.00 0.00 1 0 0 1 0
## 2 1.75 1.75 2 0 0 1 0
## 3 0.75 1.25 3 0 0 0 0
## 4 0.00 0.00 4 0 0 0 1
## 5 3.00 3.00 5 1 0 0 0
## 6 0.00 0.00 6 0 0 0 0
## treatment_4 treatment_5
## 1 0 0
## 2 0 0
## 3 0 1
## 4 0 0
## 5 0 0
## 6 0 1
```

1. Understand the experiment + pick an estimand

(1a) Identify the treatment variable and the control condition

In the dataset there is a variable that indicates which experimental condition (frame) each respondent was assigned to (including a control condition).

- Create an R object called `cond` that is a **factor** version of that variable.
- Print a frequency table of the conditions (counts per condition).

Hint: use `table()` and look at `names(dat)` / `head(dat)`.

```
# your code here
```

(1b) Define the outcome Y as the post-test measure (`post_test`) and define the pre-test measure as a covariate.

Briefly describe (1–2 sentences) what higher values of Y mean, and report the range (min/max) of Y.

```
# your code here
```

(1c) Define a single treatment contrast you will study

For the rest of the assignment, focus on ONE contrast:

- choose one framing condition (call it “treatment”)
- compare it to the control condition

Create a new data frame called `df` that keeps only observations assigned those two conditions. Then in `df`, create:

- `W` = 1 if in your chosen frame, 0 if in control
- `Y` = post-test score
- `pre` = pre-test score

Report the sample size in treatment and control.

```
# your code here
```

2. Estimate treatment effects

(2a) Difference in means

Compute the unadjusted difference in means post-test score between treatment and control, using `mean()` on the relevant subsets of `df$Y`.

$$\widehat{ATE} = \bar{Y}_{W=1} - \bar{Y}_{W=0}.$$

Save it as an object called `ate_hat_unadj`.

```
# your code here
```

(2b) Regression estimate

Estimate the treatment effect by regressing Y on W. What is your estimated treatment effect? Is it the same as in 2a? Why or why not.

```
# your code here
```

(2c) Regression controlling for pre-test score

Estimate the treatment effect controlling for pre-test score by running:

$$Y_i = \alpha + \tau W_i + \gamma pre_i + \varepsilon_i.$$

Save the estimated coefficient on W as an object called `ate_hat_adj`. Is it the same as the unadjusted estimates? Why or why not?

```
# your code here
```

3. Inference

(3a) Inference for the unadjusted estimate

Use a two-sample t-test to get a 95% CI and a (two-sided) p-value for the unadjusted effect. Report both and interpret the p-value in one sentence.

```
# your code here
```

(3b) Inference for the adjusted estimate

Use the regression output from (2b) to report: - the estimated effect (coefficient on W) - its standard error - a 95% CI - a p-value

Interpret the p-value in one sentence.

```
# your code here
```

(3c) Compare

In 3–5 sentences: compare the unadjusted and adjusted estimates. Do they differ? Why might controlling for pre-test change the estimate and/or the SE?

```
# your answer here
```

4. Randomization inference (permutation test)

In class we discussed randomization inference as a way to compute a p-value using the random assignment mechanism.

(4a) Unadjusted: write a function to get a placebo ATE

Write a function `ri_once_unadj(df)` that: - creates a permuted treatment assignment `newW` by shuffling `df$W` (without replacement) - returns the difference in means under `newW` (using `df$Y`)

(Return a single number.)

```
# your code here
```

(4b) Generate a randomization distribution (unadjusted)

Use `replicate()` to run `ri_once_unadj(df)` 1000 times and save the output as `ri_dist_unadj`. Do NOT print all 1000 values.

```
# your code here
```

(4c) Randomization-inference p-value (unadjusted)

Compute the two-sided p-value:

$$p = \Pr(|\widehat{ATE}^{\text{placebo}}| \geq |\widehat{ATE}^{\text{obs}}|).$$

Report it and compare it to the t-test p-value from (3a) (1–3 sentences).

```
# your code here
```

(4d) Adjusted: placebo distribution for the regression coefficient

Write a function `ri_once_adj(df)` that: - permutes `W` to create `newW` - runs the regression `Y ~ newW + pre` - returns the coefficient on `newW`

```
# your code here
```

(4e) Adjusted: RI p-value

Use `replicate()` to build `ri_dist_adj` (1000 draws), then compute the two-sided RI p-value for `ate_hat_adj`. Compare this RI p-value to the regression p-value from (3b).

```
# your code here
```

5. Power and MDE thinking (simulation)

Now pretend you are planning a follow-up study comparing the SAME frame vs control, and you plan to analyze post-test controlling for pre-test.

(5a) Estimate “noise” after controlling for pre-test

One simple way to approximate the remaining noise is: - fit a regression of `Y` on `pre` in the control group only - take the standard deviation of the residuals

Let `sigma_hat` be that residual standard deviation. Report `sigma_hat`.

```
# your code here
```

(5b) Pick an effect size you want to be able to detect

Choose ONE: - Option A: Use your observed adjusted effect size `ate_hat_adj` as the target effect size. - Option B: Choose a smaller “smallest effect size of interest” (SESOI) and justify it in 1–2 sentences.

Save your chosen target effect size as `delta`.

```
# your code here
```

(5c) Power curve

Write a short simulation that approximates power for a two-sample difference-in-means test ($\alpha = 0.05$, two-sided) as a function of sample size per arm.

Suggested approach (one way): - Choose sample sizes per arm, e.g. `n_grid <- seq(50, 1000, by = 50)` - For each `n` in the grid: - Simulate outcomes for control: `Y0 ~ Normal(0, sigma_hat)` - Simulate outcomes for treatment: `Y1 ~ Normal(delta, sigma_hat)` - Run a t-test and record whether $p < 0.05$ - Repeat (e.g., 500 times) and compute the fraction “significant” = estimated power

Make a plot of power vs `n`, and report the smallest `n` where power is at least 0.80.

```
# your code here
```

(5d) One-paragraph takeaway

Based on your power curve, answer: - Is the effect size you're targeting realistic/substantively meaningful? - What sample size would you recommend for a follow-up study? - What assumptions is your power calculation making that might not hold in practice?

```
# your answer here
```